PRIFYSGOL ABERYSTWYTH UNIVERSITY

Department of Computer Science Demonstrator Training October 2025

Jim Finnis

Web site

My Aber website has this information on it, and more (including a timesheet template)

https://users.aber.ac.uk/jcf12/teaching/demonstrating/landing/

Introduction

- Thanks. Really.
- Covered today:
  - Equality, Dignity and Inclusion
  - Health, Safety and Employment
  - How to Demonstrate

Dignity and Respect

- Equality and Diversity – equal respect for all:

https://www.aber.ac.uk/en/cs/equality-and-diversity/

- Dignity and respect policy
- Disabilities, learning differences and anxieties are not always visible
- Be kind and patient
- Help people towards understanding rather than show what you know
- Respect personal space (!)
- You are representing the University

Indignity and Disrespect!

- (pause the recording, Jim)
- What's the worst thing a demonstrator has ever done for you?

Dignity and Respect

- And what's the best?

(remember to start the recording again, Jim)

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Discrimination

- Gender balance in STEM is skewed
- Compsci in particular:
  - Not many women
  - Quite a few trans folk
- Do not **discriminate**

Everyone is just a student. Treat them all with exactly the same kindness and professionalism.

Discrimination

Do not treat people differently on the basis of

- Gender
- Gender identity
- Sexual orientation
- Race or nationality
- Age
- Anything else

Differently means worse **or better**!

DO NOT

- Hang around near particular students, it can make them uncomfortable
- Send students DMs / emails
- Give some students more time than others
- Ask students out during the session (!)

We hold you to a high standard of behaviour.

You are employees of the university, just as we are.

We are trusting you do this professionally.

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Name badges

- You will get a name badge.
- Wear your name badge.
- You will be able to get them from compsci reception

(you can guess why this has become necessary. Sorry.
I don't like it either.)

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Health, Safety and Employment

Emergencies

- You are not alone and responsible for everything – ask for help
- We try to have more experienced demonstrators available, and the lecturer will always be available
- Fire
- Illness – look for first aiders
- Disruptiveness – seek help from lecturing staff

Emergency contacts

It is very unlikely you'll need these numbers, but:

- In emergencies, call 2424 on an internal phone (01970 622424 on a mobile) for CompSci reception.
- In extreme cases, call 2649 for Security at the porter's lodge.

Practicalities

- Register on Aberworks
- Right to work checks
- Apply for role
- Check email regularly
  - May need to change sessions
  - May need substitutes due to illness etc.
- Payment by monthly timesheet…

Payment

- Monthly timesheet template from website
  https://users.aber.ac.uk/jcf12/teaching/demonstrating/landing/
- Complete timesheet once you've finished the teaching for a particular month
- Put only sessions for that month on the sheet!
- Get the timesheet items signed by your module coordinators, and email it as an attachment to Academic Operations.
  - Add "AberWorks Demonstrator: Computer Science" to the subject!
  - Do NOT send a link to a OneDrive document, send the actual timesheet as an attachment
- You will be paid at the end of the next month

Payment

- Get timesheets in promptly – don't save them up (this can cause Tax/NI problems)
- At Christmas get it in before you leave
- Deadlines are on the website (or will be soon)

Admin contacts

- Payment/timesheet issues:
  - Academic Operations acadops@aber.ac.uk
  - Add “AberWorks Demonstrator: Computer Science” to the subject (it’s a shared address)
- Scheduling and employment:
  - Carys Pike cyl@aber.ac.uk
- Other issues:
  - Jim Finnis (me): jcf12@aber.ac.uk

- We also have a Discord. I’ll send out invites by email.

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How to Demonstrate

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All modules are different

- I've tried to cover everything but will have missed things
- If in doubt, talk to the module coordinator

Responsibilities

- Best ability is availability – **be on time**
- **Be prepared** for the class content
- **Be proactive** – don't wait to be asked
- Be ready to interact with people
- Focus for the length of the class
- Don't come if you are unwell – but **let us know**
- Respect confidentiality
- Declare conflicts of interest

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A summary

- Read the worksheet in advance!
- You don’t have to know everything, you can get help from the lecturer
- Help students to work out the solution a step at a time
- Ask them to describe the problem as precisely as they can
- Don’t do the work for them – easy to do that incrementally
- Needs to be their work, their way
- Be aware of plagiarism
- Try to be available to all students
- Be relaxed

Preparation

- Read the worksheet in advance
- Revise the language and system
- Revise how to use the debugger (if there is one)
- Try to refamiliarize yourself with common error messages in the compiler/IDE
- Be relaxed, calm and approachable

Above all

- **Don’t be worried that you might not know the answer to the students’ questions…**

Not knowing

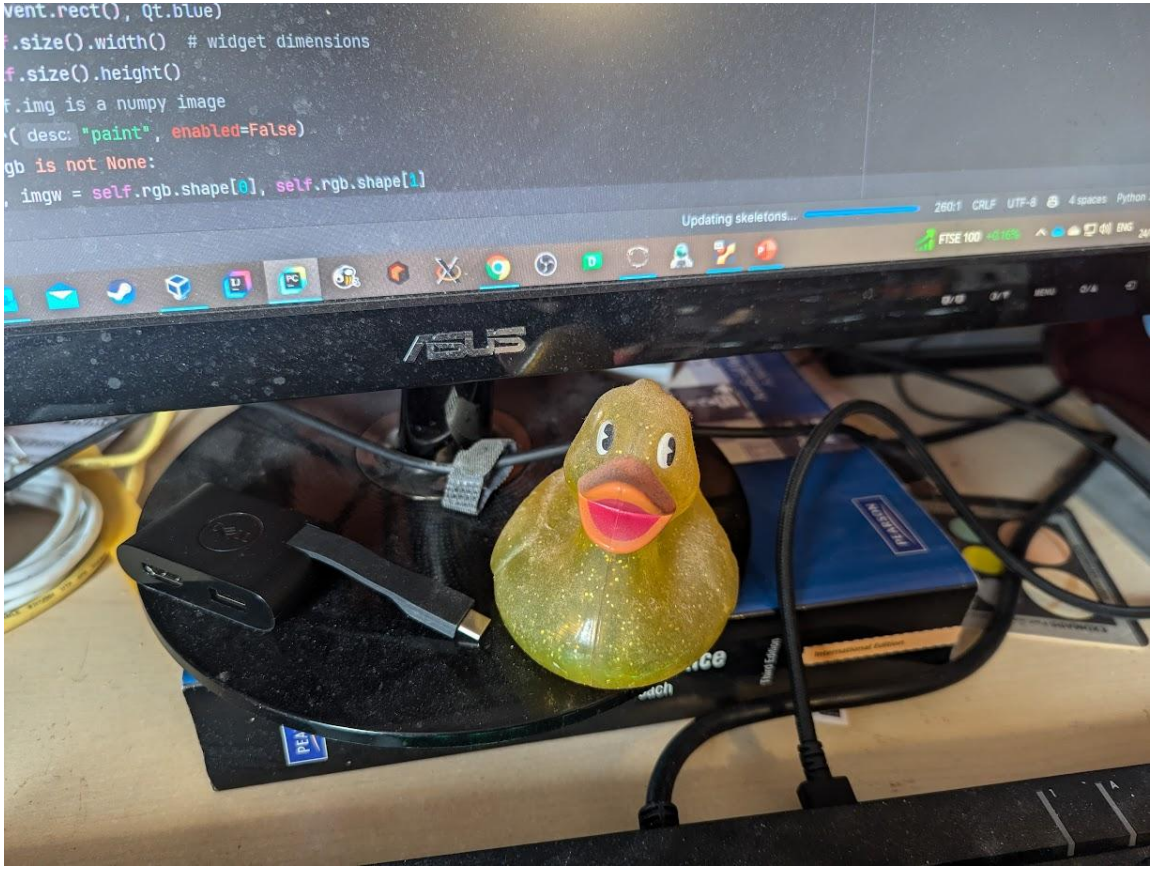
- It’s OK not know the answer to a student’s question
- It’s often better!
- **“I don’t know, but let’s find out together.”**
- Help them find out the answer themselves
- Or let them help you figure it out so they can see your thought process
- Don’t just let them watch passively

Remember – you are expected to get help from the lecturer when stuck!

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Rubber duck debugging

For when the question is “why doesn’t it work?”



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Rubber duck debugging

- **Be their rubber duck**
- Make the student explain their code to you line by line
- Explaining something **in detail** helps you see problems

"Many programmers have had the experience of explaining a problem to someone else, possibly even to someone who knows nothing about programming, and then hitting upon the solution in the process of explaining the problem. In describing what the code is supposed to do and observing what it actually does, any incongruity between these two becomes apparent."

[https://en.wikipedia.org/wiki/Rubber_duck_debugging]

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Oh no

- If their whole concept is flawed get them to tell you their thinking
- Do not just delete their code or be dismissive of it
- **Never, ever say their code is rubbish…**

Don’t lose patience

- If you think a student can’t do it, and you show it, they will believe you. Don’t even think it.
- It is very easy to lose patience and become dismissive. This makes students “disengage.”
- This can sometimes be a real challenge.

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Unacceptable Academic Practice

- All assessed work **must be the student's own** (unless stated otherwise)
- It's a fine line between getting help from a fellow student and copying their code.
- It's a fine line between getting hints from websites or AI and copying code.
- Even if unassessed work is being copied
  - the copier won't learn anything
  - and will become reliant on copying work

Unacceptable Academic Practice

- It’s OK for students to ask a fellow student a question
- It’s not OK for students to work together (because usually one student is actually doing the work)
- Keep an eye on students who appear to be working too closely together.
- Particularly watch for this when assignments are being worked on

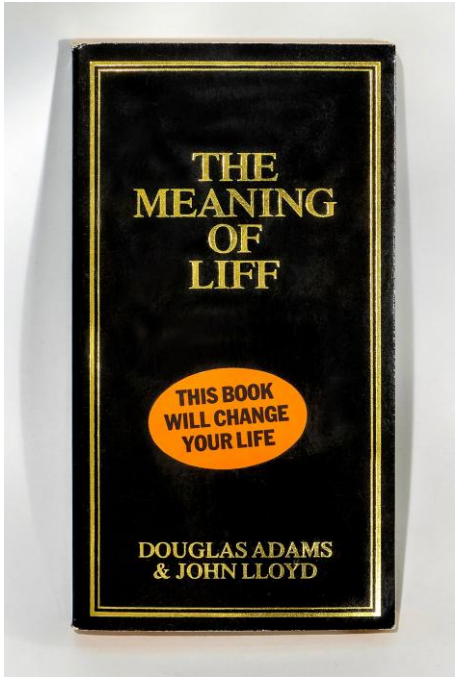
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"AI"

- (in the form of Generative AI / Large Language Models)
- If you see a student using an LLM make sure
  - They are aware that LLMs aren't a database of facts, they're a statistical model of the most likely way to complete a prompt given a big, blurry snapshot of the internet and loads of books.
  - The internet is full of lies. The LLM doesn't know the difference.
  - The blurry nature of the snapshot means it's very far from perfect.
- They should be able to understand and explain what the LLM is telling them.
- **Otherwise, they aren't learning.**
- Blackboard course on generative AI and using it ethically:
  - https://blackboard.aber.ac.uk/ultra/organizations/_54998_1/outline

Kent (adj.)

Politely determined not to help despite a violent urge to the contrary. *Kent* expressions are seen on the faces of people who are good at something watching someone else who can't do it at all.



- *The Meaning of Liff* (Douglas Adams & John Lloyd)

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Good for you too!

- This is a great opportunity for you to learn
  - Technical stuff
  - How to explain things

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End of session

- Try to make sure student work is reasonable before sign-off
- You need to know what the requirements are
- Feedback any issues to module coordinator

Please turn up.

- If you don’t come to a session you’re supposed to attend, it puts an unfair load on the other demonstrators and the lecturer.
- Students might not get seen. They might start to fall behind.
- If you can’t come, **tell us in good time.**

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Finally

Thank you.

We are really delighted that you all want to demonstrate and help other students. You can have a wonderful effect on what people will learn and how they will feel about computing and their prospects for the future. You are role models and have really helpful expertise to share.

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Any questions?

